

Control Structures & Functions

1. Control Structures: If-Elif-Else

Decision-making is anticipation of conditions occurring while execution of the program and specifying actions taken according to the conditions.

Python uses if, elif (else if), and else statements. Indentation (whitespace at the beginning of a line) is used to define scope in the code.

2. Loops in Python

- **while loop:** With the while loop we can execute a set of statements as long as a condition is true.
- **for loop:** A for loop is used for iterating over a sequence (that is either a list, a tuple, a dictionary, a set, or a string).
- **range() function:** To loop through a set of code a specified number of times, we can use the range() function. It returns a sequence of numbers.
- **break/continue:** 'break' stops the loop entirely. 'continue' stops the current iteration and continues with the next.

3. Defining Functions

A function is a block of code which only runs when it is called. You can pass data, known as parameters, into a function. A function can return data as a result.

In Python a function is defined using the **def** keyword.

Example: `def my_function(name): return f"Hello {name}"`

4. Parameters, Arguments and Return Values

Information can be passed into functions as arguments.

- **Default Parameter Value:** You can define a default value for a parameter.
- **Arbitrary Arguments (*args):** If you do not know how many arguments will be passed, add a * before the parameter name.
- **Keyword Arguments (**kwargs):** If you do not know how many keyword arguments will be passed, add two asterisks: **.

To let a function return a value, use the **return** statement.

5. Lambda Functions and Scope

A lambda function is a small anonymous function. It can take any number of arguments, but can only have one expression.

Syntax: `lambda arguments : expression`

Scope: A variable is only available from inside the region it is created. This is called scope. (Local vs Global scope).